Determinants: Definitions

The Key Fact

Fact. *A square matrix is invertible if and only if its determinant is not zero.*

The 2×2 Case

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

We use both ***det*** and vertical lines to indicate determinant.

Permutation Matrices and Signs

Defn. *A* ***permutation matrix*** *is a square matrix that contains only* 0*'s and* 1*'s with exactly one* 1 *in each row and column.*

Defn. *The* ***sign*** *of a generalized permutation matrix is* $(-1)^k$*, where* k *is the number of row interchanges needed to change the matrix to be diagonal.*

Definition of Determinant

Defn. *The determinant of matrix A: construct all possible permutation matrices P and for each, multiply relevant entries of A together then by sign of P, and sum the results.*

For example:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \text{ has perm matrices } \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \ \& \ \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

The former has positive sign with transversal ad; the latter has negative sign with transversal bc. So we get $ad - bc$.

Some Consequences

Fact. *If matrix A has an all-zero row or column, then* $\det A = 0$.

Fact. *The determinant of a triangular matrix is the product of the diagonal entries.*

In particular, the determinant of the identity matrix I is 1.

Summary

A permutation matrix is square matrix with exactly one 1 in each row and column and all other entries zero. Its sign is $(-1)^k$, where k is number of row interchanges needed to change it to diagonal.

The determinant of a matrix is defined by: construct all possible permutation matrices and for each, multiply the relevant entries together then by the sign, and then sum the results.

Summary (cont)

The determinant of $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is $ad - bc$. The determinant of a triangular matrix is the product of the diagonal entries. A square matrix is invertible if and only if its determinant is not zero.